similarities between actual or possible systems and the hypothesised global workspace, separately from the range of modules in the system or the details of neurobiological implementation, and thus develop a clearer sense of the possible functional kinds in this area.

Advocates of GWT have argued that the global workspace facilitates a range of functions in humans and other animals (Baars 1988, Shanahan 2010). These include making it possible for modules to exert ongoing control over others for the duration of a task (e.g. in the case of searching for a face in a crowd), and dealing with novel stimuli by broadcasting information about them, thus putting the system in a position to learn the most effective response. Global broadcast and the capacity to sustain a representation over time, while using it to process incoming stimuli, are necessary for these functions. Because the global workspace requires that information from different modules is represented in a common "language", it also makes it possible to learn and generate crossmodal analogies (VanRullen & Kanai 2021, Goyal et al. 2022). A particularly sophisticated and notable possible function of the global workspace is "System 2 thought", which involves executing strategies for complex tasks in which the workspace facilitates extended and controlled interactions between modules (Kahneman 2011, VanRullen & Kanai 2021, Goyal & Bengio 2022). For example, planning a dinner party may involve engaging in an extended process, controlled by this objective, of investigative actions (looking to see what is in the fridge), calls to episodic memory, imagination in various modalities (how the food will taste, how difficult it will be to cook, how the guests will interact), evaluation and decision-making. In this case, according to the theory, the workspace would maintain a representation of the goal, and perhaps compressed summaries of interim conclusions, and would pass queries and responses between modules.

We argue that GWT can be expressed in four conditions of progressively increasing strength. Systems that meet more of these conditions possess more aspects of the full global workspace architecture and are, therefore, better candidates for consciousness.

The first condition is possessing specialised systems which can perform tasks in parallel. We call these systems "modules", but they need not be modules in the demanding sense set out by Fodor (1983); they need not be informationally encapsulated or use dedicated components of the architecture with functions assigned prior to training. Mashour et al.'s recent statement of the global neuronal workspace hypothesis claims only that modules in which unconscious processing takes place are localised and specialised, and that they process "specific perceptual, motor, memory and evaluative information" (2020, p. 777). It may be that having more independent and differentiated modules makes a system a better candidate for consciousness, but GWT is most plausibly interpreted as claiming that what matters for consciousness is the process that integrates the modules, rather than their exact characteristics. The first indicator we draw from this theory is, therefore:

GWT-1: Multiple specialised systems capable of operating in parallel (modules)

Building on this, a core condition of GWT is the existence of a bottleneck in information flow through the system: the capacity of the workspace must be smaller than the collective capacity of the modules which feed into it. Having a limited capacity workspace enables modules to share information efficiently, in contrast to schemes involving pairwise interactions such as Transformers, which become expensive with scale (Goyal et al. 2022, Jaegle et al. 2021a). The bottleneck also forces the system to learn useful, low-dimensional, multimodal representations (Bengio 2017,

Goyal & Bengio 2022). With the bottleneck comes a requirement for an attention mechanism that selects information from the modules for representation in the workspace. This yields our second indicator:

GWT-2: Limited capacity workspace, entailing a bottleneck in information flow and a selective attention mechanism

A further core condition is that information in the workspace is globally broadcast, meaning that it is available to all modules. The two conditions we have seen so far are not enough to ensure that ongoing interaction between modules is possible, or that information in the workspace is available to multiple output modules which can use it for different tasks. Our third indicator is, therefore:

GWT-3: Global broadcast: availability of information in the workspace to all modules

This entails that all modules must be able to take inputs from the global workspace, including those modules which process inputs to the system as a whole. The first two conditions can be satisfied by wholly feedforward systems which have multiple input modules, feeding into a limited-capacity workspace, from which information then flows on to one or more output modules. But this new condition entails that information must also flow back from the workspace to the input modules, influencing their processing. In turn, this means that the input modules must be (algorithmically) recurrent—and thus provides further justification for indicator RPT-1—although output modules, which map workspace states to behaviour, need not be recurrent.

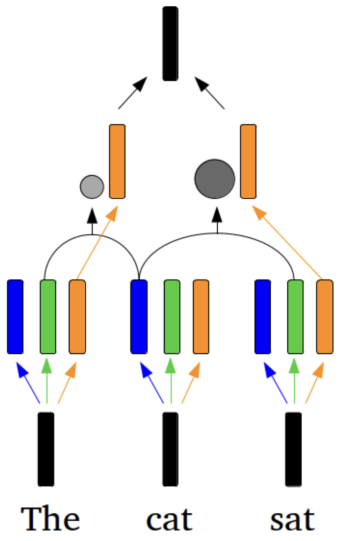
Finally, for the workspace to facilitate ongoing, controlled interactions between modules it must have one further feature. This is that the selection mechanism that determines information uptake from the modules must be sensitive to the state of the system, as well as to new inputs. That is, the system must implement a form of "top-down attention" as well as "bottom-up attention". This allows representations in the workspace itself or in other modules to affect which information is selected from each module. State-dependent selection can be readily implemented by systems that meet GWT-3 because global broadcast entails that information flows from the workspace to the modules. Generating controlled, functional interactions between modules, however, will require that the system as a whole is suitably trained. Our fourth indicator is:

GWT-4: State-dependent attention, giving rise to the capacity to use the workspace to query modules in succession to perform complex tasks

Compared to other scientific theories of consciousness, many more proposals have been made for the implementation of GWT in artificial systems (e.g. Franklin & Graesser 1999, Shanahan 2006, Bao et al. 2020). We discuss implementations of GWT, together with other theories, in section 3.1.

Box 3: Attention in neuroscience and in AI

The fields of neuroscience and machine learning each have their own distinct concepts of attention (Lindsay 2020). In machine learning, several different forms of attention have been developed, but at present, the most common is "self-attention" (Vaswani et al. 2023). This is the mechanism at the heart of Transformer networks, which power large language models.



In self-attention, representations of elements of an input sequence (for example, words in a sentence) are allowed to interact multiplicatively. Specifically, each **word representation**, that is given as a vector is transformed into three new vectors: a query, key, and value. The query vector of one word is multiplied by the key vectors of all other words to determine a weighting for each of these words. This weighting is applied to the value vectors of these words; the sum of these weighted value vectors forms the new **representation** of the word. This process is done in parallel for all words.

"Cross-attention" follows a similar formula but allows the query to be generated from one set of representations and the key and values to come from another (self-attention and cross-attention are both forms of "key-query attention"). This can be helpful, for example, in translation networks that use the words of the sentence being generated in the target language to guide attention toward the appropriate words of the sentence in the original language.

Key-query attention has only loose connections to how attention is conceptualised in neuroscience. Similar to self-attention, gain modulation (wherein attention multiplicatively scales neural activity) has been found in many neural systems (Treue & Trujillo 1999, Reynolds & Heeger 2009). However, this attentional modulation is frequently thought to arise from recurrent top-down connections, not from the parallel processing of concurrent inputs (Noudoost et al. 2010, Bichot et al. 2015). Previous versions of attention in machine learning have relied on recurrent processing, and in this way could be considered more similar to biological attention (Mnih et al. 2014, Bahdanau et al. 2014). However, it should be noted that there are many different flavors of attention within neuroscience and the underlying neural mechanisms may vary across them. Therefore, saying definitively which forms of artificial attention are closest to biological attention in general, is not straightforward.

Insofar as different theories of consciousness depend on recurrent processing or other specific components of the attention mechanism, self-attention may not be sufficient to form the basis of artificial consciousness. For example, there is nothing akin to the binary ignition process in global workspace theory in self-attention, as attention is implemented as a graded weighting of inputs. There is also no built-in model of the attention process on top of attention itself, as required in attention schema theory.

2.3 Higher-Order Theories

2.3.1 Introduction to higher-order theories

The core claim of higher-order theories of consciousness is helpfully distilled by Brown et al. (2019):

> The basic idea . . . is that conscious experiences entail some kind of minimal inner awareness of one's ongoing mental functioning, and this is due to the first-order state being in some ways monitored or meta-represented by a relevant higher-order representation. (p. 755)

Higher-order theories are distinguished from others by the emphasis that they place on the idea that for a mental state to be conscious the subject must be aware of being in that mental state, and the way in which they propose to account for this awareness. This is accounted for by an appeal to higher-order representation, a concept with a very specific meaning. Higher-order representations are ones that represent something about *other representations*, whereas first-order representations are ones that represent something about the (non-representational) world. This distinction can be applied to mental states. For example, a visual representation of a red apple is a first-order mental state, and a belief that one has a representation of a red apple is a higher-order mental state.

Higher-order theories have long been advocated by philosophers (Carruthers & Gennaro 2020, Rosenthal 2005). One of the main motivations for the view is the so-called "simple argument" (Lycan 2001): if a mental state is conscious, the subject is aware that they are in that state; being aware of something involves representing it; so consciousness requires higher-order representation of one's own mental states. The substantive commitment of this argument is that there is a single sense of "awareness" of mental states on which both premises are true—which is both weak enough that consciousness entails awareness of mental states, and strong enough that this awareness entails higher-order representation. In the last two decades, higher-order theories have been elaborated, refined and tested by neuroscientists, and influenced by new experimental methods and ideas from the study of metacognition, signal detection theory, and the theory of predictive processing.

A variety of higher-order theories have been proposed, which describe distinct forms of monitoring or meta-representation, and imply different conditions for consciousness (Brown et al. 2019). They include: several philosophical theories, including higher-order thought theory (Rosenthal 2005) and higher-order representation of a representation theory (Brown 2015); the self-organising meta-representational account (Cleeremans et al. 2020); higher-order state space theory (Fleming 2020); and perceptual reality monitoring theory (Lau 2019, 2022, Michel forthcoming). We will concentrate on perceptual reality monitoring theory (PRM) and to some degree also the closely-related higher-order state space theory (HOSS). These are both recent computational theories based on extensive assessments of neuroscientific evidence.

The core claim of PRM is that consciousness depends on a mechanism for distinguishing meaningful activity in perceptual systems from noise. There are multiple possible sources of neural activity in perceptual systems. This activity could be caused by perceptible stimuli in the environment; it could be sustained after these stimuli have passed; it could be generated top-down

through expectations, imagination, dreaming or episodic memory; or it could be due to random noise. PRM claims that a "reality monitoring" mechanism, which operates automatically, is used to discriminate between these different kinds of activity and assess the reliability of first-order representations. Perceptual representations are conscious when they are identified as reliable, or in other words, as being sufficiently different from noise.

Meanwhile, HOSS makes the similar claim that "awareness is a higher-order state in a generative model of perceptual contents" (Fleming 2020, p. 2). This higher-order state, which is the product of a metacognitive inference, signals the probability that some particular content is represented in the perceptual system. This is presented as a theory of the basis of awareness reports (i.e. reports of the form "I am/not aware of X"), but Fleming suggests that higher-order awareness states are necessary for consciousness.

2.3.2 Computational HOTs and GWT

Computational higher-order theories are sometimes grouped together with GWT as "global" theories, in opposition to "local" theories such as RPT (Michel & Doerig 2022). Like GWT, higher-order theories such as PRM claim that cognitive functions supported by the prefrontal cortex play an important role in consciousness. As such, the evidence reviewed above in favour of the involvement of the PFC in consciousness supports PRM as well as GWT. PRM also claims, again like GWT, that "consciousness is the *gating mechanism* by which perception impacts cognition; it selects what perceptual information should directly influence our rational thinking" (Lau 2022, p. 159). Lau (2022) endorses the existence of global broadcast as a phenomenon in the brain, and also affirms that it is *related* to consciousness: when representations are conscious, "global broadcast and access" of that representation "are *likely* to happen" (p. 159).

However, higher-order theorists reject the claim that broadcast in the global workspace is necessary and sufficient for consciousness. Notably, according to higher-order theories, unconscious representations can be encoded in the global workspace, which implies that a representation might be unconscious and yet available for high-level cognitive processes, such as reasoning. Higher-order theories and GWT make distinct predictions, and advocates of computational HOTs appeal to experiments testing these predictions as providing important evidence in favour of their view.

In one such experiment, Lau and Passingham (2006) conducted a visual discrimination task under a range of different masking conditions and asked participants to press a key to indicate whether they had seen or merely guessed the shape of the stimulus in each trial. They identified two different masking conditions in which participants' ability to discriminate between stimuli was at the same level, but differed in how likely they were to report having seen the stimulus. Higher-order theorists interpret this result as showing that there can be a difference in conscious perception of a stimulus without a corresponding difference in task performance, and claim that this result is inconsistent with a prediction of GWT (Lau & Rosenthal 2011). This purported prediction of GWT is that differences in consciousness should entail differences in task performance, because—according to GWT—consciousness makes information available to a wide range of cognitive functions, useful across a wide range of tasks. Furthermore, according to GWT, ignition leading to global broadcast is necessary and sufficient for consciousness, and ignition depends on the same factors which affect visual task performance, such as signal strength and attention.